

The 13th International Symposium on Frontiers in Ambient and Mobile Systems
(FAMS 2023)

March 15–17, 2023, Leuven, Belgium

Shapley Values for Explaining the Black Box Nature of Machine Learning Model Clustering

Mouad Louhichi^{a,*}, Redwane Nesmaoui^a, Marwan Mbarek^a, Mohamed Lazaar^a

^aENSIAS, Mohamed V University in Rabat, Morocco

Abstract

Machine learning (ML) models are becoming increasingly complex. In fact, a sophisticated model (XGBoost boosting or deep learning) generally leads to more accurate predictions than a simple model (linear regression or decision tree). There is therefore a trade-off between the performance of a model and its interpretability: what a model gains in performance, it loses in interpretability (and vice versa), where interpretability is the ability for a human to understand the reasons for a model's decision. However, explaining the predictions made by machine learning models aims at computing and interpreting the importance of features. To achieve this, game theory has recently gained attention for better understanding the similarity between group members. In this paper, we use SHAP (SHapley Additive exPlanations), which is a method based on cooperative game theory, to analyze and evaluate the properties of each group. More importantly, we rely k-means PCA and Light gbm classifier to improve the data preparation before grouping the features into multiple clusters. The simulation results prove the importance of shapley value in creating an accurate and meaningful representation of each cluster.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Conference Program Chairs

Keywords: Game theory; Shapley Value; Machine learning; Clustering; Dimensionality reduction; Explainability; Visualization

1. Introduction

The discussion around Data Science is giving an increasing amount of weight to the importance of the explainability of algorithms. We are aware of the strength that algorithms possess, and they are able to provide assistance to us in a variety of activities, including the prediction of prices, the categorization of documents, and the suggestion of movies.[1] However, more and more people are questioning the ethics, bias, and validity of this prediction[?]. Interpretability in machine learning helps ensure objectivity in decision-making by identifying and addressing biases in the training data. It also promotes resilience by highlighting any potential issues that may affect the prediction

* Corresponding author. Tel.: +212 664 80 9073

E-mail address: mouad.louhichi@um5.ac.ma

and ensures that the results are based on significant factors. In certain situations, it may be more beneficial to use deterministic algorithms with clear principles rather than machine learning techniques that may be facing challenges or have specific regulatory requirements to consider.[2]

In order to accomplish these objectives, a new discipline known as XAI (Explainable Artificial Intelligence) has arisen. The objective of this field is to develop algorithms that are both powerful and explainable. In order to better explain non-transparent algorithms, several different frameworks have been developed. There is a general and unified way to do that in a cooperative game with transferable utility (TU).[3]. In contrast to the methods employed by other researchers, the axiomatic aspects of TU games are used in order to provide support for TU game solution ideas. The Shapley value is the most well-known example of a solution of this kind. It satisfies a variety of criteria, including those pertaining to efficiency, justice, and symmetry. Shapley values are an original and equitable method for determining how much each participant should get from a cooperative game. Recently, there have been a variety of Shapley-value-based techniques for determining which input characteristics affected the prediction of a machine learning (ML) model.[18] SHAP, KernelSHAP, TreeSHAP QII, and IME are among the most significant ones. The most significant step in applying the Shapley values technique to ML models is to build up a cooperative game in which the input characteristics will act as the players and the model prediction will serve as the reward.

Accordingly, the paper is organized as follows. In [section 1](#), we introduce the system to be studied. In [section 2](#), we present related works that points out that Shapley values have been extensively used in the context of eXplainable Artificial Intelligence (XAI) to provide feature attributions as explanations. In [section 3](#), we focus on cooperative game theory and specifically on Shapley value by presenting its mathematical formula and its various features. Then, we present in [section 4](#) the simulation results of our clustering-based machine learning, starting by presenting our database, preparing our tree-based clustering model in order to use Shapley value to explain each cluster.

2. Related Works

Shapley values have been widely used in the context of Explainable AI (XAI) to provide feature attributions as explanations [3]. However, since computing Shapley values is intractable [6], various approximations have been used [4]. These techniques implement the Shapley value method from cooperative game theory, concentrating on the axiomatic motivation of Shapley values and efficient ways for calculating them. In general, only a few works attempt to demonstrate how tiny variations in the underlying game formulations of current approaches. One such work is The Explanation Game: Explaining Machine Learning Models Using Shapley Values [2], which is an example of such an attempt. [7], Or Joia et al [9] who use principal component analysis to carry out an easy matrix decomposition in order to determine the features that are the most important, and who then show feature names as word clouds inside each cluster location.

3. Shapley value based game theory

3.1. Cooperative Game theory

In the literature of Game theory, a "Game" is any situation with multiple decision-makers, each of whom wants to get the best results possible[14]. The best choice will depend on what other people do. The game shows who the players are, what they like, what strategies they can use, and how these strategies affect the outcome [10]. Game Theory tries to use math to describe these situations and figure out what would happen if everyone acted logically.

3.2. Shapley value

The Shapley value is a game theory approach that allocates the total payoff of a cooperative game fairly among game's players. Recently, various applications of this method have been revealed in the field of explaining ML models.

Shapley values give a framework for fairly assigning payoffs to individual participants in a game that may be played alone or in coalitions.

As is common practice, let's note N the set of all players, n the total number of players, and $v : 2^N \rightarrow \mathbb{R}$ the function such that $v(S)$ is the payoff received by the coalition of players S

Based on Shapley's value, the fair amount the player should receive is defined as follows [12]

$$\phi_i(v) := \frac{1}{n} \sum_{S \subset N \setminus \{i\}} \binom{n-1}{|S|}^{-1} \Delta v_i(S)$$

where $\Delta v_i(S) := v(S \cup \{i\}) - v(S)$. A way to understand $\phi_i(v)$ is that, if player i joins a coalition S , it is only fair that she asks as attribution for her contribution $v(S \cup \{i\}) - v(S)$.

In order to avoid giving individuals a bonus or penalty for entering a coalition early or late, we consider the average of all possible permutations that could lead to coalition formation.

The Shapley value is the only payment allocation rule that meets the four characteristics listed below.

1. **Efficiency:** Individual awards are equal to the payoff of the coalition composed of all participants minus the value of the empty coalition.: $\sum_{i=1}^n \phi_i(v) = v(N) - v(\emptyset)$
2. **Symmetry:** If two players i and j always have the same impact in any coalition they are added to and that contains neither, then they should be attributed the same payoff. That is, if $v(S \cup \{i\}) = v(S \cup \{j\})$ for every S such that $i \notin S$ and $j \notin S$, then $\phi_i(v) = \phi_j(v)$.
3. **Linearity:** Running two attributions one after the other would produce the same result as running one attribution with the combined payoff: $\forall v, w, i, \quad \phi_i(v + w) = \phi_i(v) + \phi_i(w)$
4. **Null player :** A player that does not contribute when added to any coalition should not be attributed any payoff: if $\forall S, v(S \cup \{i\}) = v(S)$ then $\phi_i(v) = 0$ [5].

3.2.1. Shapley Values For Predictive Modeling

Shapley values provide an adaptable framework for understanding the smallest influence of a feature while creating a predictive model. In a game that involves predictive modeling, the players who work together to solve problems are referred to as features. The use of multiple features by a model is analogous to a coalition of players coming together to play the game. Depending on what we intend a certain feature to be, the unique rules of the game and the payoff function v will vary [15].

4. Simulation Results

The SHAP library is used to explain the characteristics of the clusters in our dataset. By understanding which features had the most influence on the clustering result and how each feature contributes to the predicted cluster for each sample, we can present a clear and understandable explanation of the clusters. In this context, a sample refers to a single item in the dataset, such as a single wine in a collection of wine data. By using the SHAP library to calculate Shapley values for each feature for each sample, we can understand how the features of each sample contribute to its predicted cluster, and how those features compare to the features of other samples in the same cluster..

4.1. Data Set

The dataset is related to white variant of the Portuguese "Vinho Verde" wine. These dataset can be viewed as classification or regression tasks. The classes are ordered and not balanced (e.g. there are many more normal wines than excellent or poor ones). Outlier detection algorithms could be used to detect the few excellent or poor wines. [8]

4.2. Data Preparation

In order to explain clusters using the Shapley value, we must first scale our data, reduce the feature dimensions of the dataset, and construct a multiclass classifier to predict the cluster from the features.

Table 1. The dataset is represented as a 2D Matrix in a python numpy array

Id	fixed acidity	...	volatile acidity
1	7.0	...	0.12
2	2.4	...	0.831
3	3.07	...	0.12
4	1.4	...	0.94
...	...	...	...

4.2.1. Scaling

Feature scaling is a preprocessing step that is used to standardize or normalize the input data. It is often necessary when working with machine learning algorithms because the algorithms can be sensitive to the scale of the input features. When the range of values in a feature is very different from that of other features, it can dominate the model and skew the results. By scaling the features to a common level, we can ensure that all features are weighted equally and that the model is trained properly. In our case, we have features with widely varying domains. For example, the mean value of the 'fixed acidity' feature is 6.854788, while the mean value of the 'volatile acidity' feature is 0.278241. This indicates that these features are on different scales, so we need to scale them to a common level before we can use them together in our machine learning model.

4.2.2. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a popular unsupervised machine learning technique that is used to reduce the number of input variables in the training dataset. This technique comes under dimensionality reduction technique. Perform principal component analysis to get an idea of the dimensionality of the wine dataset. Perform dimension reduction to two dimensions with PCA to be able to draw all the wines in one graph

4.2.3. Kmeans clustering

The k-means algorithm is a widely used and popular method for clustering in pattern recognition and machine learning. While it is an unsupervised learning technique, the k-means algorithm and its extensions can be affected by the initialization with a predetermined minimum number of clusters. [13] To determine the optimal number of clusters for k-means clustering, we looped from 2 to 11 clusters and calculated the within-cluster variation (inertia), the silhouette score, and the Davies-Bouldin index for each result. The elbow method, which seeks the value of k after which the clustering quality improves only marginally, was used to determine the number of clusters. However, it is important to note that the k-means algorithm is non-deterministic, so we should consider running multiple iterations in order to obtain more robust results. In this case, we obtained a high Davies-Bouldin score and a low silhouette score for k=3, and therefore chose to cluster into three clusters.

4.2.4. Light gbm classifier

A gradient boosting system called Light GBM makes use of a tree-based learning technique. While other algorithms grow trees horizontally, Light GBM grows trees vertically, which translates to Light GBM growing trees leaf-wise while other algorithms grow levels-wise. The leaf with the greatest delta loss will be chosen to grow. A gradient boosting system called Light GBM makes use of a tree-based learning technique. While other algorithms grow trees horizontally, Light GBM grows trees vertically, which translates to Light GBM growing trees leaf-wise while other algorithms grow levels-wise. The leaf with the greatest delta loss will be chosen to grow. We may still find it challenging to explain the clusters. Now, we will build a multiclass classifier to predict the cluster based on the features. Next, we shall describe the clusters using the Shapley values.

4.3. Explain clusters with Shapley values

we use the SHAP library to understand the characteristics of the clusters in our dataset. By examining which features had the most influence on the clustering result and how each feature contributes to the predicted cluster for each sample, we can present a clear and understandable explanation of the clusters.

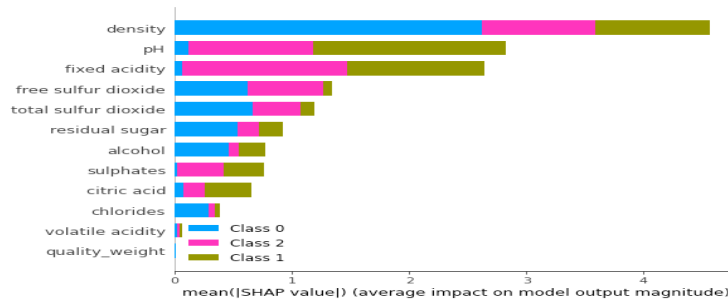


Fig. 1. average impact on model output magnitude

The above graph shows the mean absolute SHAP value for each feature. This plot provides an overview of which features had the most influence on the clustering result, either individually or in combination with other features, we can see that the average impact of Shapley values indicates that feature density has the highest impact on the cluster prediction model. and if we look at the Shapley value per cluster. The feature's acidity pH and fixed acidity have an impact only on clusters 1 and 2 but almost none on cluster 0. Next, we used SHAP dependence plots to visualize the relationship between each feature and the predicted cluster for each sample. These plots show the effect of a single feature on the predicted cluster, taking into account the values of the other features. By examining the dependence plots for each feature, we were able to understand which features were most important in determining the membership of a sample in a particular cluster

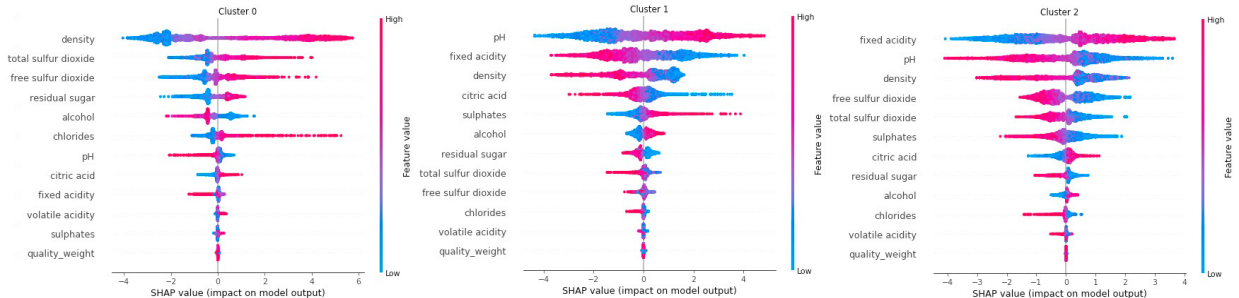


Fig. 2. Explain clusters

- **Cluster 0:** it seems that the SHAP dependence plot for cluster 0 shows the relationship between the features "density", "total sulfur dioxide", and "free sulfur dioxide" and the predicted cluster for each sample. The plot likely indicates that wines with high values for these features are more likely to be assigned to cluster 0. Overall, the SHAP dependence plot for cluster 0 suggests that the features "density", "total sulfur dioxide", and "free sulfur dioxide" have a significant impact on the predicted cluster for each sample, and that wines with high values for these features are more likely to be assigned to cluster 0. This summary of the plot provides an overview of the characteristics that define cluster 0 and how they relate to the quality of the wine.
- **Cluster 1:** Explain cluster 1: the plot shows the relationship between the features "pH", "fixed acidity", "density", "citric acid", and "sulphates" and the predicted cluster for each sample. The plot likely indicates that wines with high values for "pH" and "sulphates" and low values for "fixed acidity", "density", and "citric acid" are more likely to be assigned to cluster 1. Altogether, the SHAP dependence plot for cluster 1 suggests that the features "pH", "fixed acidity", "density", "citric acid", and "sulphates" have a significant impact on the predicted cluster for each sample, and that wines with high values for "pH" and "sulphates" and low values for "fixed acidity", "density", and "citric acid" are more likely to be assigned to cluster 1. This summary of the plot provides an overview of the characteristics that define cluster 1 and how they relate to the quality of the wine.
- **Cluster 2:** The SHAP dependence plot for cluster 2 shows the relationship between the features "fixed acidity", "pH", and "density" and the predicted cluster for each sample. The plot likely indicates that wines with high

values for "fixed acidity" and low values for "pH" and "density" are more likely to be assigned to cluster. In general the SHAP dependence plot for cluster 2 suggests that the features "fixed acidity", "pH", and "density" have a significant impact on the predicted cluster for each sample, and that wines with high values for "fixed acidity" and low values for "pH" and "density" are more likely to be assigned to cluster 2. This summary of the plot provides an overview of the characteristics that define cluster 2 and how they relate to the quality of the wine."

5. Conclusion

Shapley values are a tool for explaining the predictions made by a machine learning model. They allow you to understand the specific contribution of each feature to the prediction for a given sample, and to quantify the impact of that feature on the prediction. Unlike some other interpretability techniques, Shapley values do not rely on any particular assumptions and take into account the interactions between the features when determining their importance. The SHAP library is a popular tool for estimating Shapley values in Python, and it offers a range of algorithms and visualizations to help you understand and interpret the results. However, it is important to be aware of the limitations of Shapley values, both in terms of the mathematical definition and the estimation process, in order to ensure that you are using them appropriately and interpreting the results correctly. One challenge with using Shapley values is that they do not provide any direct measure of the accuracy or reliability of the explanations they generate, so it is important to consider other sources of information and validation when interpreting the results.

References

- [1] Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., ... Lee, S. I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature machine intelligence*, 2(1), 56-67.
- [2] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), 206-215.
- [3] Lundberg, S. M. and Lee, S.-I. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.
- [4] Sundararajan, M. and Najmi, A. The many shapley values for model explanation. In *International conference on machine learning*, pp. 9269–9278. PMLR, 2020.
- [5] Yves-Laurent Kom Samo 2022, Feature Engineering With Game Theory: Beyond SHAP values, <https://blog.kxy.ai/feature-engineering-with-game-theory-beyond-shap>
- [6] Bang, S., Xie, P., Lee, H., Wu, W., Xing, E. (2021, May). Explaining a black-box by using a deep variational information bottleneck approach. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 35, No. 13, pp. 11396-11404).
- [7] Merrick, R., Taly, A. (2020). The Explanation Game: Explaining Machine Learning Models Using Shapley Values https://doi.org/10.1007/978-3-030-57321-8_2
- [8] P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. Modeling wine preferences by data mining from physicochemical properties. In *Decision Support Systems*, Elsevier, 47(4):547-553. ISSN: 0167-9236.
- [9] Joia, P., Petronetto, F., Nonato, L., 2015. Uncovering representative groups in multidimensional projections. *CGF* 34, 281–290
- [10] Rozemberczki, B., Watson, L., Bayer, P., Yang, H., Kiss, O., Nilsson, S., Sarkar, R., 2022. The Shapley Value in Machine Learning. <https://doi.org/10.48550/arXiv.2001.10167>
- [11] Merrick, L., Taly, A. (2019). The explanation game: Explaining machine learning models with cooperative game theory. *arXiv preprint arXiv:1909.08128*.
- [12] Lahiri, A., Alipour, K., Adeli, E., Salimi, B. (2022). Combining counterfactuals with shapley values to explain image models. *arXiv preprint arXiv:2206.07087*.
- [13] Pelegrina, G. D., Siraj, S. (2022). Shapley value-based approaches to explain the robustness of classifiers in machine learning. *arXiv preprint arXiv:2209.04254*.
- [14] Serrano, R. (2007). Cooperative games: Core and Shapley value (No. 2007-11). Working Paper.
- [15] Sundararajan, M., Najmi, A. (2020, November). The many Shapley values for model explanation. In *International conference on machine learning* (pp. 9269-9278). PMLR.
- [16] allón-Ballesteros, A., Chen, C. (2020). Explainable AI: Using Shapley value to explain complex anomaly detection ML-based systems. *Machine learning and artificial intelligence*, 332, 152.
- [17] Liu, Y., Chen, C., Zhang, X., Xie, S. (2020, October). Shapley values and meta-explanations for probabilistic graphical model inference. In *Proceedings of the 29th ACM International Conference on Information Knowledge Management* (pp. 945-954).
- [18] NAVARRO, Constanza L. Andaur, DAMEN, Johanna AA, TAKADA, Toshihiko, et al. Risk of bias in studies on prediction models developed using supervised machine learning techniques: systematic review. *bmj*, 2021, vol. 375.